

Workshop Sheet - Week 1

Reading and writing a Lean proof. InfoView displays your hypotheses and current goal. Use this tactic reference alongside the worked examples and syntax hints in the Lean file.

<code>intro h</code>	Assume the premise of an implication.
<code>exact h</code>	Submit <code>h</code> as a proof of the current goal.
<code>apply h</code>	Apply the result <code>h</code> .
<code>rw [h]</code>	Rewrite using an equality.
<code>rfl</code>	Prove an equality whose two sides are equal by definition.
<code>use k</code>	Choose <code>k</code> as a witness for an existence goal.
<code>cases h</code>	Split a hypothesis into its possible cases.
<code>obtain ⟨k, hk⟩ := h</code>	Extract a witness and its property from an existence hypothesis.
<code>have h : P := by ...</code>	State and prove an intermediate claim.
<code>by_contra h</code>	Assume the goal is false (call this <code>h</code>), to derive a contradiction.
<code>simp</code>	Simplify the goal using standard rewrite rules.
<code>norm_num</code>	Normalize numbers: simplify and prove numerical facts.
<code>positivity</code>	Prove that an expression is positive, nonnegative or nonzero.
<code>linarith</code>	Combine linear equations and inequalities ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$).
<code>nlinarith</code>	Deduce consequences of polynomial equalities and inequalities.

Indentation groups proof steps; `·` starts a separate proof branch.

1. Tactic practice.

- Prove that any proposition P implies itself.
- Given $k \in \mathbb{N}$, prove that there exists $m \in \mathbb{N}$ with $m = k$.
- For integers a, b , deduce $a + 1 = b + 1$ from $a = b$.
- For $m \in \mathbb{N}$, prove $(m + 1) - m = 1$.
- For $n \in \mathbb{N}$, if $n = 2$ or $n = 3$, prove $n^2 \leq 9$.
- For $x, y \in \mathbb{R}$ with $x + y \leq 7$ and $3 \leq y$, prove $x \leq 4$.
- For $x \in \mathbb{R}$, prove $x^2 + 1 > 0$.
- For $x \in \mathbb{R}$ with $x^2 \leq 4$, prove $x \leq 2$.
- Verify that $\frac{3}{4} + \frac{1}{6} = \frac{11}{12}$ in $\mathbb{Q}$.

2. Infinitely many primes.

- For each $n \in \mathbb{N}$, prove that there exists $m \in \mathbb{N}$ with $n < m$.
- For $d, a, b \in \mathbb{N}$, prove that $d \mid a$ and $d \mid b$ imply $d \mid a + b$.
- For $a, b, c \in \mathbb{N}$, prove that $a \mid b$ and $b \mid c$ imply $a \mid c$.
- Prove that $n! + 1$ has a prime divisor.
Finding facts. Find factorial positivity and prime-divisor existence in [Loogle](#). Inspect the results with `#check`.
- Suppose p is prime, $p \leq n$ and $p \mid n! + 1$. Derive a contradiction.
Hint. Search for divisibility of a factorial and divisibility of a difference.
- Formalize the statement and proof of the following theorem.
Theorem. *There are infinitely many prime numbers.*

3. The irrationality of $\sqrt{2}$.

- For integers a, b satisfying $a^2 = 2b^2$, exhibit a witness proving $2 \mid a^2$.
- For $a \in \mathbb{Z}$, deduce $2 \mid a$ from $2 \mid a^2$.
Hint. Search for a result about a prime dividing a product, then consider the two cases.
- For integers a, b satisfying $a^2 = 2b^2$, prove that a is even.
- Substitute $a = 2k$ into $a^2 = 2b^2$ and deduce $b^2 = 2k^2$.
Hint. Rewrite the hypothesis, then use `nlinarith`.
- Show that coprime natural numbers a, b cannot both be even.
Hint. A common divisor divides the greatest common divisor.
- Formalize the statement and proof of the following theorem.
Theorem. *The number $\sqrt{2}$ is irrational.*